

2003 EXT 1 July Common

Question 1 (10 marks)

- a) i) Write an expression for $\cos(A + B)$ (1)
ii) Hence find the exact value of $\cos 105^\circ$ (2)
- b) Find the coordinates of P (x, y) that divides the interval AB externally in the ratio 5:1 given A (0,2) and B (3,0) (2)
- c) Find the acute angle between the lines $x - 2y - 6 = 0$ and $y = -3x + 4$ to the nearest minute. (2)
- d) Prove that the line $3x - y - 10 = 0$ is a tangent to the circle $x^2 + y^2 = 10$ (3)

Question 2 (8 marks) Start a new page

- a) Find i) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ (1)
ii) $\lim_{x \rightarrow \infty} \frac{2x}{x + 2}$ (2)
- b) Solve $2 \sin \frac{x}{2} = \cos \frac{x}{2}$ for $0^\circ \leq x \leq 360^\circ$ (3)
- c) The minute and hour hands of a clock are 9cm and 6cm in length respectively. Find the distance between the ends of the hands when the time is 5 o'clock (to the nearest mm). (2)

Question 3 (10 marks) start a new page

- a) Using $\tan \frac{\theta}{2} = t$ write $\frac{1}{2} \cot \frac{\theta}{2} - \cot \theta$ in terms of t in simplest form (2)
- b) Differentiate with respect to x, $y = \frac{x}{3\sqrt{x}}$ (2)
- c) Prove $\sin(\frac{\pi}{4} + A) - \sin(\frac{\pi}{4} - A) = \sqrt{2} \sin A$ (3)
- d) Solve $\sin 2x + \cos x = 0$ for $0^\circ \leq x \leq 360^\circ$ (3)

Question 4 (10 marks) Start a new page

a) Differentiate

i) $\frac{1}{4-x^2}$ (2)

ii) $\frac{1-x^2}{1+x^2}$ (2)

b) i) Express $\sqrt{3} \cos x - \sin x$ in the form $A \cos(x + \theta)$ where $0^\circ \leq \theta \leq 90^\circ$ (2)

ii) Hence or otherwise solve $\sqrt{3} \cos x - \sin x = 1$ where $0^\circ \leq x \leq 360^\circ$ (2)

c) If $y = \sqrt{r^2 - x^2}$ where r is a constant, show that $\frac{dy}{dx} = \frac{-x}{y}$ (2)

Question 5 (9 marks) Start a new page

a) Find the coordinates of the points on the curve $y = 2x^3 - 9x^2 + 27$ where the tangent is parallel to the x axis. (3)

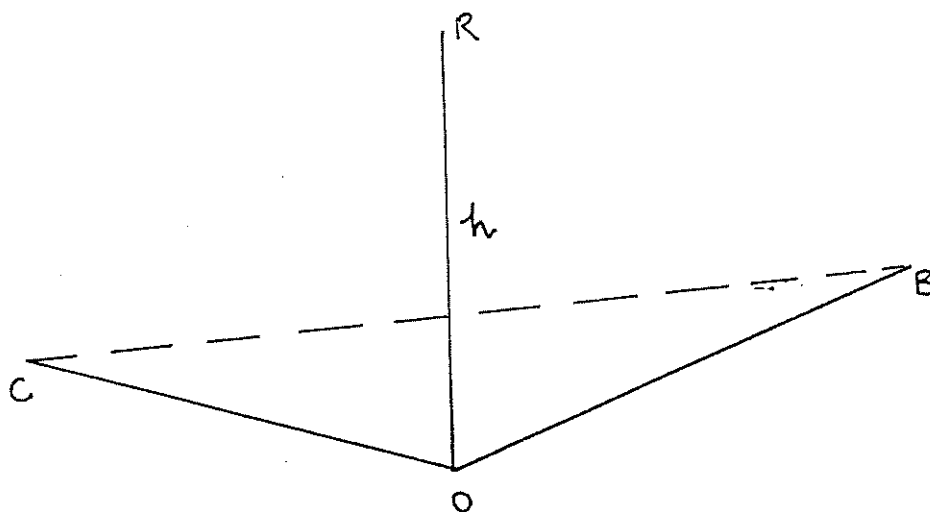
b) Given $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$, solve the equation $\cos 3\theta + \cos \theta = 0$ for $-180^\circ \leq \theta \leq 180^\circ$ (3)

c) i) For the function $f(x) = x\sqrt{2-x}$ find $f'(x)$ (2)

ii) Hence solve $f'(x) = 0$ (1)

Question 6 (10 marks) Start a new page

- a) From a tower OR , the bearings of two points C and B , on the same level ground as the base of the tower, are 300° and 015° respectively. The angles of elevation to the top of the tower are 11° and 22° from C and B respectively. BC is 150m . The tower OR has height h metres.



- i) Redraw the above diagram showing all relevant information. (1)
 - ii) Find $\angle BOC$ (2)
 - iii) Find OC and OB in terms of h (3)
 - iv) Find the height of the tower to the nearest metre. (3)
- b) i) If $\sin(x + \theta) = k \sin(x - \theta)$ prove
 $(k - 1) \tan x = (k + 1) \tan \theta$ (2)
- ii) Hence solve $\sin(x + 20^\circ) = 2 \sin(x - 20^\circ)$ if $0^\circ \leq x \leq 360^\circ$ (2)

QUESTION 1

i) $\cos(A+B) = \cos A \cos B - \sin A \sin B$

$\cos 105^\circ = \cos(60^\circ + 45^\circ)$
 $= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$
 $= \frac{1}{2} \cdot \frac{\sqrt{2}}{2} - \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2}$

$= \frac{1-\sqrt{3}}{2\sqrt{2}}$

$A(2, 2)$ $B(3, 0)$

$\left(\frac{(1 \times 0) + (-5 \times 2)}{-4}, \frac{(1 \times 2) + (-5 \times 0)}{-4} \right)$

$P\left(\frac{15}{4}, -\frac{1}{2}\right)$

$x - 2y - 6 = 0 \therefore y = \frac{x-6}{2}$

let $m_1 = \frac{1}{2}$

$y = -3x + 4$ let $m_2 = -3$

$\tan \theta = \left| \frac{\frac{1}{2} - (-3)}{1 + \frac{1}{2} \times (-3)} \right|$

$\tan \theta = 7$

$\therefore \theta = 81^\circ 52'$ (to nearest min)

circle $x^2 + y^2 = 10$ has

centre $(0, 0)$ radius $= \sqrt{10}$

perp dist $(0, 0)$ to $3x - y - 10 = 0$

$r = \frac{|-10|}{\sqrt{9+1}}$

$r = \frac{10}{\sqrt{10}}$ rationalise

$r = \sqrt{10} \therefore$ since perp dist. is equal to radius $\Rightarrow$ tangent

QUESTION 2

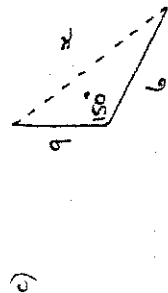
a) i) $\lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)} = 4$

ii) $\lim_{x \rightarrow \infty} \frac{2x}{x+2} = 2$

b) $2 \sin \frac{x}{2} = \cos \frac{x}{2}$

$2 \tan \frac{x}{2} = 1$
 $\tan \frac{x}{2} = \frac{1}{2}$
 $\frac{x}{2} = 26^\circ 34', 206^\circ 34'$

$x = 53^\circ 8'$ in domain



$x^2 = a^2 + b^2 - 2 \times a \times b \cos 150^\circ$

$\therefore x = 14.5 \text{ cm (1 d.p.)}$

or 14.5 mm

QUESTION 3

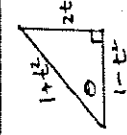
a) $\frac{1}{2} \cot \frac{\theta}{2} - \cot \theta$

$= \frac{1}{2} \cdot \frac{1}{t} - \frac{(1-t^2)}{2t}$

$= \frac{1-1+t^2}{2t}$

$= \frac{t^2}{2t}$

$= \frac{t}{2}$



SOLUTIONS

2003 EXT 1

Question 4

a) i) $\frac{d}{dx} (4-x^2)^{-1} = -1 \times -2x (4-x^2)^{-2}$
 $= \frac{2x}{(4-x^2)^2}$

ii) use quotient rule

$u = 1-x^2 \quad v = 1+x^2$

$u' = -2x \quad v' = 2x$

$\frac{dy}{dx} = \frac{-2x(1+x^2) - 2x(1-x^2)}{(1-x^2)^2}$

$= \frac{-2x - 2x^3 - 2x + 2x^3}{(1-x^2)^2}$

$= \frac{-4x}{(1-x^2)^2}$

b) i) $A = \sqrt{3+1} = 2$
 $2\left(\frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x\right)$

required form

$2(\cos x \cos \theta - \sin x \sin \theta)$
 $\therefore \cos \theta = \frac{\sqrt{3}}{2} \quad \sin \theta = \frac{1}{2}$

$\therefore \theta = 30^\circ$

$\therefore \sqrt{3} \cos x - \sin x = 2 \cos(x+30^\circ)$

ii) $2 \cos(x+30^\circ) = 1$

$\cos(x+30^\circ) = \frac{1}{2}$

$x+30^\circ = 60^\circ, 300^\circ$

$x = 30^\circ, 270^\circ$

c) $y = (r^2 - x^2)^{-1/2}$

$\therefore \frac{dy}{dx} = \frac{1}{2} \times -2x (r^2 - x^2)^{-3/2}$

$= \frac{-x}{\sqrt{r^2 - x^2}}$

since $\frac{-x}{y} = \frac{-x}{\sqrt{r^2 - x^2}}$

$\therefore \frac{dy}{dx} = \frac{-x}{y}$

b) $y = \frac{1}{3} x \cdot x^{1/2}$

$\therefore y = \frac{1}{3} x^{3/2}$

$\frac{dy}{dx} = \frac{1}{3} \times \frac{3}{2} x^{1/2}$

$= \frac{1}{2} \sqrt{x}$

c) Prove $\sin\left(\frac{\pi}{4} + A\right) - \sin\left(\frac{\pi}{4} - A\right) = \sqrt{2} \sin A$
 LHS $= \sin \frac{\pi}{4} \cos A + \cos \frac{\pi}{4} \sin A - \left(\sin \frac{\pi}{4} \cos A - \cos \frac{\pi}{4} \sin A\right)$

$= \frac{1}{\sqrt{2}} \cos A + \frac{1}{\sqrt{2}} \sin A - \frac{1}{\sqrt{2}} \cos A + \frac{1}{\sqrt{2}} \sin A$

$= \frac{2}{\sqrt{2}} \sin A$

$= \frac{2 \times \sqrt{2}}{\sqrt{2}} \sin A$

$= \frac{2\sqrt{2}}{2} \sin A$

$= \sqrt{2} \sin A$

$= \text{RHS}$

d) $\sin 2x + \cos x = 0$
 $2 \sin x \cos x + \cos x = 0$

$\cos x (2 \sin x + 1) = 0$

$\cos x = 0 \quad \sin x = -\frac{1}{2}$

$\therefore x = 90^\circ, 270^\circ \quad \text{acute } x = 30^\circ$
 $\therefore x = 210^\circ, 330^\circ$

